

Assessing the Education–Job Skill Gap: A Survey-Based Study of Curriculum Relevance, Practical Training, and Employability Readiness

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Abstract—The aim of this report is to find out how wide the gap is between what education teaches and what employers actually want, and what the people inside that system think should be done about it. An 80-item questionnaire was prepared in Google Forms and shared online, and 100 valid responses were collected from students and early-career respondents in technical, business, humanities and vocational fields. The gap is widely felt: 75% agreed that a real gap exists, but only 27% felt their education had prepared them for the career they want. Most respondents blamed their own institutions rather than the outside world: outdated curriculum (34%) and lack of practical training (32%) made up two-thirds of the answers, against 14% for fast technological change. Practical, job-specific skills were the weakest area (37%), and 46% said more internships would help the most. The report concludes that the problem lies in how education is delivered, and recommends compulsory internships, regular curriculum review with industry input, and direct teaching of digital and communication skills.

Index Terms—education skill gap, employability, curriculum relevance, practical training, industry–academia collaboration, graduate readiness

I. INTRODUCTION

A. Background Information

A degree used to be proof that a person could do the job. That is no longer true in the same way. Employers now hire for what a candidate can actually do, and they often say that new graduates know the theory but cannot yet apply it to real work [1]. This difference between what education supplies and what the job market demands is what is called the *education skill gap*.

The cost is heavy on both sides. For a graduate it means a longer job search, lower pay, or work that has nothing to do with the subject studied. For a country like Bangladesh it

means a poor return on the money families and the government spend on education. Local conditions make it worse: teaching methods change slowly, universities have little contact with industry, and technical and vocational training is not given much respect [2]. The pressure is also rising, because automation and artificial intelligence keep changing the tasks a job contains, and a syllabus revised once every few years cannot keep up [5].

Research on the gap is plentiful, but almost all of it looks at the problem from one side. It asks employers what they want, or it audits what institutions deliver. Far less of it asks learners themselves, and almost none asks them three linked questions at once: how big the gap feels, which skills they think were neglected, and what they would change first. That is the gap this report addresses: without those three answers, reform gets designed for students rather than with them.

B. Research Question and Hypothesis

The report therefore asks: *how wide is the education skill gap for the people living inside it, what do they believe causes it, and what do they think would close it?* From the literature above, one hypothesis was set before the responses were examined: respondents will report a wide skill gap and will blame the way their institutions teach more than they blame outside technological change.

The scope is limited: the survey records belief about preparation rather than tested skill, no employers were surveyed, and the data come from one point in time.

II. LITERATURE REVIEW

The gap is a global problem. A review of employability skills worldwide found the same complaint in country after country: what institutions teach and what recruiters hire for keep drifting apart [1]. Its main point is that the problem is rarely a shortage of knowledge. It is a shortage of the ability to use that knowledge at work.

What actually helps is less certain. A systematic review of employability programmes in higher education found that such programmes can improve both skills and job outcomes, but the benefit depends on how long the placement lasts and whether it is properly assessed [3]. Simply adding an internship rule is not enough. A separate review of information technology courses found that communication, teamwork and problem solving are still under-taught and rarely assessed, even where course documents say they matter [4]. The written syllabus and the delivered one are two different things.

Bangladesh faces the same problem with local causes. A study of graduate skill gaps in Bangladesh during the Fourth Industrial Revolution reported a large mismatch, especially in advanced technical and digital skills, and traced it to outdated teaching methods, weak links with industry, and low regard for vocational education [2]. On the technology side, analysis of artificial intelligence and labour demand found that most workers exposed to AI will not need specialist AI skills, but the tasks they do, and therefore the skills those tasks need, will change [5].

Together these studies describe the gap well, but mostly from above. This survey puts the three learner-side questions of Section I into one instrument.

III. MATERIALS AND METHODS

A. Survey Design

A descriptive, cross-sectional survey was used, and both quantitative and qualitative questions were accounted for. The quantitative items give numbers that can be averaged; the qualitative items give categories that can only be counted. Every question was closed-ended so that both could be tallied directly. The questionnaire, titled *Education Skill Gap Survey*, has 80 items in three blocks: 30 qualitative multiple-choice items on background and habits, such as “What do you think is the MAIN cause of the education skill gap?”; 30 quantitative statements rated from 1 (Strongly Disagree) to 5 (Strongly Agree), such as “Internships and practical training should be a mandatory part of every degree”; and 20 qualitative Yes/No items, such as “Have you ever used AI tools (such as ChatGPT) to learn new skills?”.

B. Medium Used

The survey was built in Google Forms and shared entirely online through Facebook Messenger, WhatsApp and email. An online form was chosen because it reaches people in different places at no cost, stores the answers in a ready format, and lets respondents answer without an interviewer present. The form is available at: <https://docs.google.com/forms/d/e/1FAIpQLSeSbaaUQuT-TuCoEpejgPkjnlmQ6MjNv5Ah0PIsTZ-kgsR4w/viewform>

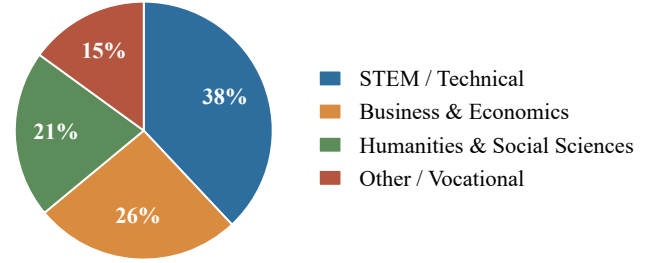


Fig. 1. Distribution of respondents by current field of study or work.

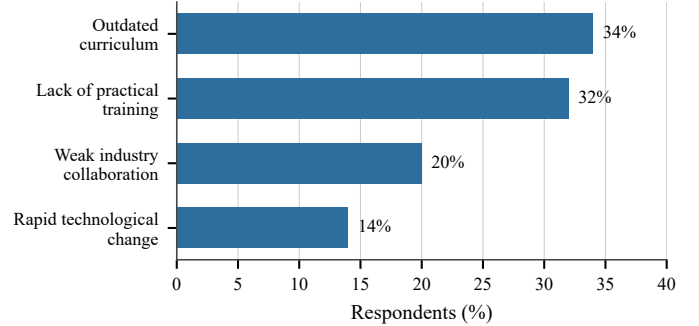


Fig. 2. Perceived main cause of the education skill gap.

C. Selection of Participants

Taking part was voluntary and anonymous; no names or institution details were collected, and respondents were told so on the form’s first screen. One hundred valid responses were kept. They came from students and early-career respondents across four broad fields, shown in Fig. 1. Since the form was shared through the authors’ own online contacts, this is a convenience sample rather than a random one.

D. Data Analysis

The qualitative answers from the multiple-choice and Yes/No blocks were counted as frequencies and percentages. The quantitative rated statements were summarised by mean and standard deviation, and each mean was read against the fixed band scale given under Table II. Both types are reported together in Section IV.

IV. RESULTS

Fig. 1 shows who answered: 38% from STEM and technical fields, 26% business and economics, 21% humanities and social sciences, and 15% other or vocational. No single field dominates, but the sample leans technical.

Fig. 2 shows what respondents blame. Outdated curriculum leads at 34%, with lack of practical training close behind at 32%. Weak industry collaboration takes 20% and rapid technological change only 14%. The two causes that point at the institution therefore account for 66% between them.

Table I gives the full multiple-choice results. Asked which skill area their education served least, respondents chose practical and job-specific skills most often at 37%, ahead of

TABLE I
KEY CATEGORICAL FINDINGS FROM THE EDUCATION SKILL GAP
SURVEY

Response option	Freq.	%
<i>Main cause of the education skill gap</i>		
Outdated curriculum	34	34
Lack of practical training	32	32
Weak industry collaboration	20	20
Rapid technological change	14	14
<i>Skill area education served least</i>		
Practical / job-specific skills	37	37
Technical / digital skills	25	25
Problem-solving / critical thinking	20	20
Communication / soft skills	18	18
<i>Best way institutions can reduce the gap</i>		
More internships and practical training	46	46
Updating the curriculum	31	31
More industry collaboration	23	23

n = 100 responses per item.

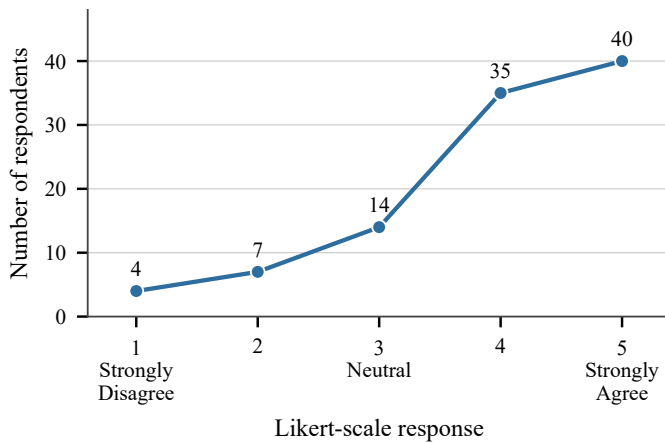


Fig. 3. Likert-response pattern on the perceived gap between education and employer skill requirements.

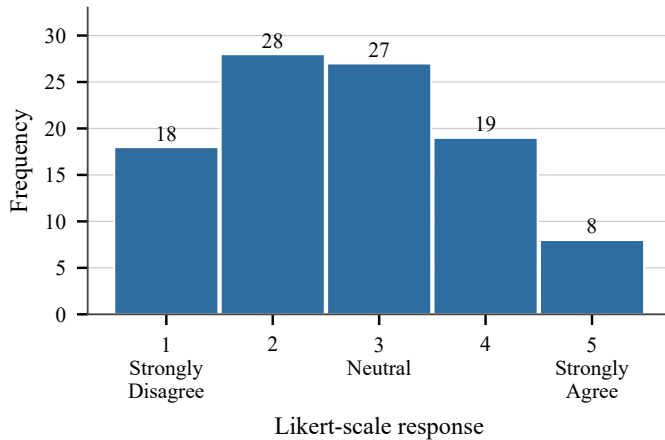


Fig. 4. Distribution of perceived career preparation provided by education.

technical and digital skills at 25%. Communication came last at 18%. Asked what institutions should do, 46% chose more internships and practical training.

Fig. 3 and Fig. 4 tell opposite stories. On the statement that a significant gap exists, the answers climb steadily towards

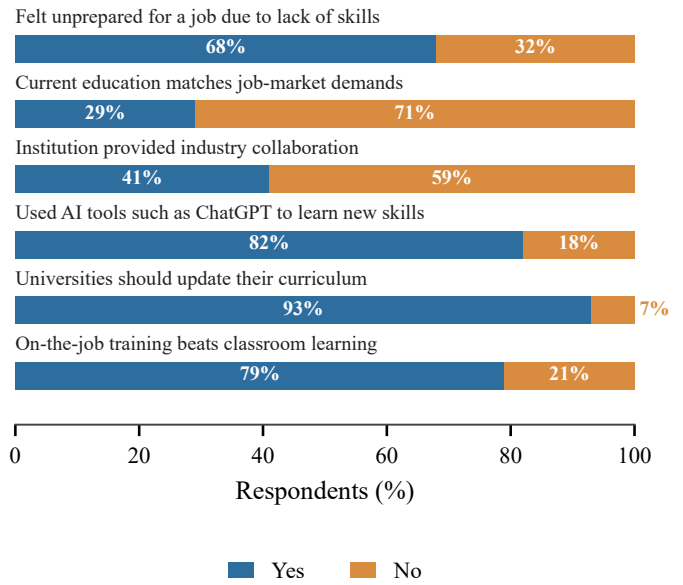


Fig. 5. Selected Yes/No findings related to education, skills, artificial intelligence, curriculum and training.

TABLE II
SUMMARY OF SELECTED LIKERT-SCALE FINDINGS

Statement (abridged)	Mean	SD	Reading
Education has prepared me for my desired career	2.71	1.20	Neutral
Significant gap between taught and required skills	4.00	1.09	Agree
Universities put theory before practical skills	3.76	1.13	Agree
Curriculum is up to date with industry needs	2.74	1.17	Neutral
Internships should be mandatory in every degree	4.29	0.95	Str. Agree
Institutions should collaborate more with industry	4.30	0.90	Str. Agree
Continuous learning is needed to stay employable	4.57	0.77	Str. Agree
Bridging the gap would improve employment rates	4.42	0.84	Str. Agree

Scale: 1.00–1.80 Strongly Disagree; 1.81–2.60 Disagree; 2.61–3.40 Neutral; 3.41–4.20 Agree; 4.21–5.00 Strongly Agree. *n* = 100. Statements are abridged; full wording is in the survey instrument.

agreement, ending at 75% agreement against 11% disagreement. On the statement that education prepared them well, the pattern reverses: only 27% agreed while 46% disagreed, and the most common answers were disagreement and neutrality.

Fig. 5 presents the Yes/No block. Sixty-eight percent had felt unprepared for a job because of missing skills, and only 29% believed their education matches what the job market wants. Just 41% had been given any industry contact by their institution, while 82% had already used AI tools such as ChatGPT to learn new skills and 79% said on-the-job training beats classroom learning. The strongest single result is that 93% want universities to update their curriculum.

Table II summarises the rated statements. The two lowest scores both concern the respondent's own institution, and both

land in the neutral band. The two middle scores describe the problem and sit in the agree band. The four highest all concern solutions and reach the strongly-agree band. One pattern runs through the whole table: as the means rise, the standard deviations fall steadily, from 1.20 down to 0.77.

V. DISCUSSION AND LIMITATIONS

The hypothesis holds on both parts. Three quarters of the respondents agreed that a real gap exists, and the statement scores 4.00, well inside the agree band. On cause, outdated curriculum and missing practical training take 66% of the blame between them, while rapid technological change, the only cause outside the institution's control, takes 14%. Respondents are not saying the world moved too fast to follow. They are saying their institutions did not move at all. The rated statements agree: the two items about their own institution sit in the neutral band, while the statement that universities put theory before practice reaches 3.76. This matches the finding that outdated teaching and weak industry links, not technology alone, drive the mismatch in Bangladesh [2].

The fix respondents want follows from that. Internships lead the remedies at 46%, the matching statement reaches 4.29, and 79% say on-the-job training beats classroom learning. The evidence supports the direction but not the confidence: such programmes do improve skills and job outcomes, yet the benefit depends on how long the placement lasts and whether it is properly assessed [3]. A compulsory but token internship would give respondents what they asked for and still fail them.

One result was not expected. Respondents ranked practical and job-specific skills as the weakest area (37%) and put communication last (18%), the reverse of what reviews of technical education report [4]. People seem to rank what they felt the absence of, and missing hands-on practice hurts at once while missing communication training only shows its cost years later. A soft-skill gap can therefore be real and still go unreported, a useful warning against setting curriculum priorities from learner opinion alone.

The answers on AI point the same way. Some 82% had used AI tools to learn new skills, yet only 14% blamed technological change for the gap. They are not treating AI as the problem; they are using it as the workaround, which fits evidence that AI changes which tasks a job contains rather than removing the need for skill [5]. The worrying part is that AI literacy is becoming a basic requirement picked up informally, so nobody is checking its quality.

Taken together the answers are consistent. The multiple-choice blame, the rated statements and the Yes/No answers all land on the same two things, curriculum and practical training. Set against 93% who want the curriculum updated and only 41% who were ever offered industry contact, the demand is clear and unmet. The wider point is that this is not mainly a problem of content. Respondents are not asking to be taught different subjects. They are asking to be taught the same subjects in a way that ends in doing something.

These findings should still be read with care. The survey measures opinion, not tested skill. The 100 respondents were

reached through the authors' own online contacts, so the sample is not representative and leans towards STEM. Rated self-reports also attract agreement with statements that sound sensible. The data come from a single point in time and no employers were surveyed, so no cause-and-effect claim can be made, and the report records what respondents believe employers want rather than what employers say.

VI. CONCLUSION AND RECOMMENDATIONS

Across 100 responses the gap is clear, and respondents put the cause inside the institution rather than outside it. They back practical training and curriculum reform almost without dissent. As this sample sees it, the problem is not getting into education. It is what education delivers once you are there.

For institutions. Make credit-bearing internships a required part of every degree, designed for real duration and proper assessment rather than for compliance, since design decides whether they work [3]. Review the curriculum on a fixed cycle with industry advisers. Move some assessment weight from memory towards applied work, and teach digital, AI and communication skills inside the syllabus.

For students. Treat regular self-directed learning as part of the job, and record it through certificates or micro-credentials so that employers can see the skill without relying on the degree alone.

For policymakers. Reward lasting industry-academia partnerships through placement subsidies and accreditation credit, and give technical and vocational education equal respect.

Future work should run a matching survey with employers in the same market, to establish whether the gaps learners feel are the ones employers actually screen for.

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